

ORIGINAL ARTICLE

A survey of anaesthetic techniques used for caesarean section in the UK in 1997

K. U. Shibli, I. F. Russell

Department of Anaesthesia, Hull Royal Infirmary, Kingston-upon-Hull, UK

SUMMARY. A prospective survey of anaesthesia for caesarean section was performed for the year 1 January to 31 December, 1997. Two hundred and fifty maternity hospitals were sent questionnaires from which 129 responses were obtained. The data provided information on anaesthesia for 60 455 caesarean sections. Overall 78% of sections were performed with regional anaesthesia: 47% single shot spinal; 22% epidural; 9% combined spinal epidural (CSE); 22% general anaesthesia. For elective caesarean sections (39% of all sections) regional anaesthesia was used for 87% of cases: 68% single shot spinal; 3% epidural; 15% CSE; 13% general anaesthesia. For emergency procedures regional anaesthesia was used for 72% of cases: 34% single shot spinal; 34% epidural; 4% CSE; 28% general anaesthesia. There was a wide range of regional anaesthesia use among the units, varying from an overall rate of 95% at one extreme to 41% at the other. Similarly, there was a wide range of conversion of regional anaesthesia to general anaesthesia, varying from 0% to 88%. Overall, 10.6% of the general anaesthetics were the result of regional to general anaesthesia conversion. © 2000 Harcourt Publishers Ltd

INTRODUCTION

There is a belief that the decline in anaesthetic-related maternal mortality is linked to the substitution of regional for general anaesthesia for caesarean section.^{1,2} This belief was not entirely supported by the findings of a previous UK survey of anaesthesia for caesarean section.³ This survey demonstrated the expected reduction in the percentage use of general anaesthesia, with the rate declining from 77% to 44% between 1982 and 1992, but there was only a modest 13% reduction in the actual number of general anaesthetics administered over this time scale, falling from 7471 (1982) to 6468 (1992). In addition, it is well recognised that the majority of deaths related to anaesthesia occur following emergency general anaesthesia^{1,4-7} but, although the rate of general anaesthesia for emergency caesarean sections fell from 85% (1982)

to 54% (1992) there was scarcely any reduction in the number of emergency general anaesthetics given in the surveyed hospitals: 4983 (1982) and 4900 (1992).³ These unexpected findings were due to a 51% increase in the total number of caesarean sections, and a 55% increase in the number of emergency sections, performed in these hospitals.

Although the data contained in the Reports on Confidential Enquiries into Maternal Deaths (CEMD)^{1,4-7} are extremely valuable there are no national denominator data available on anaesthesia for caesarean section so it is impossible to calculate the relative safety of one type of anaesthetic versus another. Since the data in the previous retrospective survey³ relate to 1982, 1987 and 1992 and anaesthetic practices continue to evolve, we felt that a prospective national survey of obstetric anaesthesia in the UK during 1997 would be worthwhile. The information obtained may then provide denominator data to assist in the interpretation of anaesthetic-related maternal deaths as detailed in the various CEMD reports.^{1,4-7}

METHODS

A database of maternity units was obtained from the Obstetric Anaesthetists' Association and data packs

Accepted October 1999

K. U. Shibli Specialist Registrar (Anaesthesia), **I. F. Russell** Consultant Anaesthetist, Department of Anaesthesia, Hull Royal Infirmary, Anlaby Road, Kingston-upon-Hull, East Yorkshire, HU14 3JU, UK

Correspondence to: I. F. Russell, Tel: +44 (0)1482 674542; Fax: +44 (0)1482 674371; E-mail: I.F.Russell@medschool.hull.ac.uk

were sent out to 250 units along with a cover letter in November 1996. The cover letter explained the nature of the survey and the period of data collection: 1 January – 31 December, 1997. The data forms (Appendix) allowed the majority of caesarean sections to be recorded by a single tick in a box. For more complex cases, such as multiple anaesthetic techniques, the order in which the anaesthetics were attempted was to be indicated by numbers in the boxes and additional comments placed in the small 'complications if any' area at the end of the line. Examples were provided on a sample sheet attached to the data forms.

Since some anaesthetists already collected the data electronically we agreed to accept their data in electronic form. On detailed examination, some of the electronic raw data and accompanying printed summary data did not correspond; double counting was occurring: if regional anaesthesia failed and was converted to general anaesthesia both techniques were counted thus inflating the total number of caesarean sections. Where the provision of raw data allowed, this double counting was corrected.

Further letters were sent out during 1997 to remind anaesthetists of the survey and to enquire about progress. In January 1998 prepaid envelopes were sent to all units and we received the data over the next 5 months, in some cases after written reminders or a phone call.

The data from individual hospitals were entered on a single line in a spreadsheet (Excel 97). When all data were entered the 'rank and percentile' macro (Excel 97) was used to provide median values, quartiles and ranges. A simple linear regression (SPSS v 7.1.5) was used to examine the relationship between the number of caesarean sections performed and the use of regional anaesthesia in individual hospitals.

RESULTS

From the 250 data packs sent out 129 responses were eventually obtained. Two hospitals on the mailing list no longer existed, six other hospitals had amalgamated into three, and two units now provided midwife-led low risk services where no caesarean sections took place. Notification was also received from two consultant anaesthetists who refused to collect any data. Eighty-four units completed the survey forms and data in electronic formats were received from a further 36 units. We therefore received data from 120 units.

Overall 22% of caesarean sections were completed with general anaesthesia, 47% with single shot spinal, 22% with epidural, and 9% with CSE (Fig. 1).

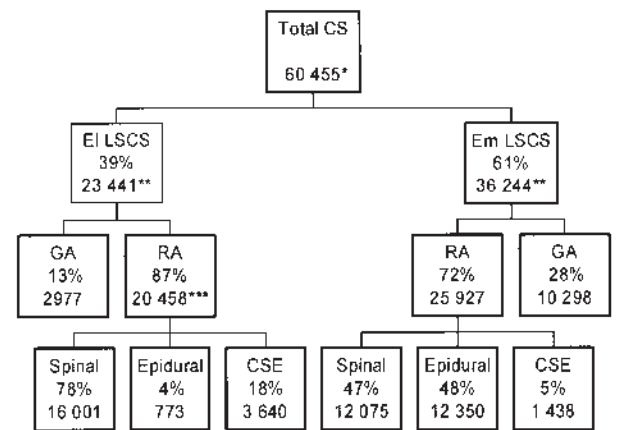


Fig. 1 Anaesthetic techniques used for all caesarean sections from 120 UK hospitals. The percentages shown for spinal, epidural and CSE relate to total number of regional anaesthetics only. CS = caesarean section; El CS = elective caesarean section; Em CS = emergency caesarean section; GA = general anaesthesia; RA = regional anaesthesia; CSE = combined spinal epidural anaesthesia. *The elective/emergency split was not shown for 770 cases (248 spinal, 262 epidurals, 210 general anaesthetics, 50 CSEs). **The anaesthetic was not specified for 6 elective and 19 emergency cases. ***The regional anaesthesia technique was not specified in 44 elective and 64 emergency cases.

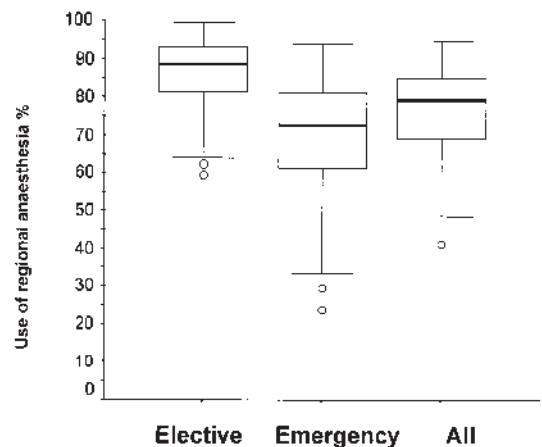


Fig. 2 Box-and-whisker plot showing the percentage use of regional anaesthesia for elective caesarean sections, emergency caesarean sections and the total number of caesarean sections in 120 obstetric units in the UK. The upper and lower edges of the box represent the 25th and 75th quartiles, the horizontal bar represents the median and the whiskers represent 1.5 times the interquartile range. The points (o) plotted individually outside the whiskers are outliers – defined as cases lying between 1.5 and 3 box heights from the nearest box edge.

Anaesthesia for elective surgery consisted of general anaesthesia 13%, single shot spinal 68%, epidural 3%, and CSE 15%; anaesthesia for emergency surgery was general anaesthesia 28%, single shot spinal 34%, epidural 34%, CSE 4%.

Regional anaesthesia rates varied widely among hospitals (Fig. 2) and a significant negative

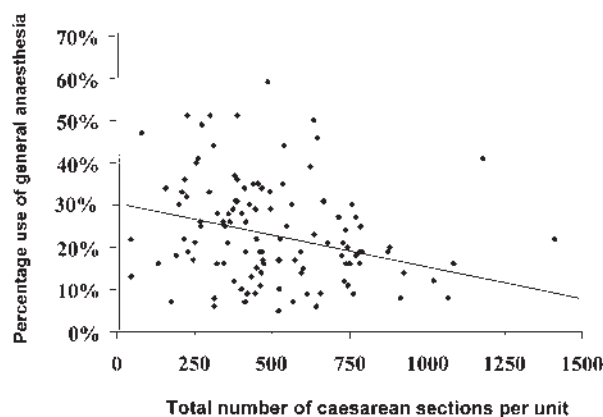


Fig. 3 Relationship between number of caesarean sections performed in each unit and the use of general anaesthesia. The regression line shows a statistically significant relationship (the Pearson correlation coefficient = -0.469 ; $P = 0.014$).

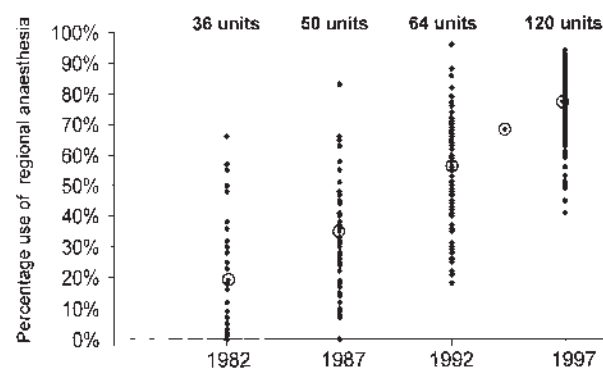


Fig. 4 Changes in the percentage use of regional anaesthesia for caesarean section between 1982 and 1997 in UK hospitals. *○ Median percentage use of regional anaesthesia in each of the years presented. *○ This point is extracted from the Department of Health Statistical Bulletin.⁷ 1982, 1987 and 1992 data are obtained from the original survey data.³

Table 1. The number of conversions of a primary regional anaesthetic technique to another technique

Primary anaesthetic technique	Total CS	Original anaesthetic technique converted to:				
		Spinal <i>n</i> (%)	Epidural <i>n</i> (%)	CSE <i>n</i> (%)	GA <i>n</i> (%)	Total <i>n</i> (%)
Spinal	28 872		41 (0.1)	5 (0.02)	503 (1.7)	549 (1.9)
Epidural	14 403	146 (1.0)		9 (0.1)	863 (6.0)	1018 (7.1)
CSE	5 230	30 (0.6)	28 (0.5)		44 (0.8)	102 (2.0)

GA = general anaesthesia.

correlation was noted between use of general anaesthesia and the total number of caesarean sections performed ($P = 0.014$) (Fig. 3). The major contribution to this statistical relationship was lower general anaesthesia rates for elective caesarean sections in units performing greater numbers of caesarean sections. There was no significant correlation between regional anaesthesia use for emergency caesarean section and either the total number of caesarean sections or the number of emergency caesarean sections performed.

Changes in regional anaesthesia rates among UK units are shown in Figure 4. The median use of regional anaesthesia for all caesarean sections rose from 20% in 1982 (quartiles 3%, 30%; range 0% to 68%) to 78% in 1997 (quartiles 69%, 85%; range 41% to 95%). The official Department of Health figure for the use of regional anaesthesia for caesarean section in 1994⁸ is also shown in Figure 4. This estimate falls precisely on the line joining the median regional anaesthesia use in 1992 with the median in 1997.

Table 1 shows the number of cases where the primary regional anaesthetic technique was converted to another anaesthetic technique. Conversions of regional anaesthesia to general anaesthesia accounted for 8% and 11.5% of the elective and emergency general anaesthetics respectively, although conversion

rates varied widely among units (Fig. 5: only units that performed successfully 50 or more of a particular regional anaesthetic technique are included in the data for that technique in Fig. 5). Of the spinals and CSEs that failed, 36% and 30% respectively were converted to general anaesthesia intraoperatively: the remaining spinals and CSEs were converted to general anaesthesia preoperatively, either because of technical problems causing a complete failure, or because the block was detected as being inadequate before surgery. Only 53% of the elective epidurals that were converted to general anaesthesia were identified as inadequate preoperatively. Among the epidurals for emergency caesarean section, in 140 cases no attempt was made to top up a functioning epidural and these accounted for 28% of the emergency epidural 'failures'. Of the 366 functioning epidurals that were used for emergency surgery and were subsequently converted to general anaesthesia, 325 (89%) were found to be inadequate after surgery started.

There were 17 reported serious complications secondary to regional anaesthesia.

- Five spinals for elective caesarean section were converted to general anaesthesia: three were described as high spinals, one patient fainted, and

Table 2. The number of elective and emergency general anaesthetics used for caesarean section and the number of failed intubations with their subsequent management

	Total number	Management of anaesthetic after a failed intubation			
		Spinal	Epidural	CSE	GA continued
Elective	2 977	2	–	–	2 (both LMA)
Emergency	10 298	6	2	–	3 (LMA, M&A, ETT)
Total	13 275	8	2	–	5

CSE = combined spinal epidural; GA = general anaesthesia; LMA = Laryngeal mask airway; M&A = Mask and Guedel airway; ETT = Endotracheal tube.

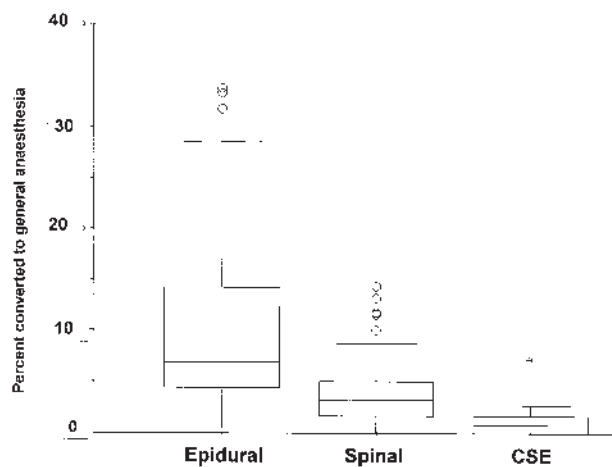


Fig. 5 Box-and-whisker plot (as in Fig. 2) showing the regional to general anaesthesia conversion rates for the various regional anaesthetic techniques (spinal, epidural and combined spinal epidural (CSE)), as used in 120 obstetric units in the UK. The points (o) plotted individually outside the whiskers are outliers (defined as cases lying between 1.5 and 3 box heights from the nearest box edge). The points (*) plotted individually outside the whiskers are extreme cases (defined as cases lying outside 3 box heights from the nearest box edge). In the epidural group there were 5 extreme values not shown at this scale in the figure. These extreme values were 45%, 45%, 71%, 72%, 88%.

one was described as inadvertent general anaesthesia. The latter occurred because thiopentone was administered instead of Augmentin after delivery of the baby.

- Six spinal for emergency surgery were converted to general anaesthesia: four were described as high spinal. Two other spinal were converted to general anaesthesia intraoperatively because of what was described as 'severe maternal bradycardia'.
- Six epidurals for emergency caesarean section were converted to general anaesthesia: five of these were described as high or total spinal and one patient had general anaesthesia induced prophylactically. This latter patient with pregnancy induced hypertension (PIH) had an epidural for labour which was topped up for emergency caesarean

section, but in the operating theatre she became irritable and it was thought she was about to have a seizure, either from PIH or from a toxic effect of local anaesthetic, so general anaesthesia was induced.

Thirty-eight potentially life-threatening complications of general anaesthesia were reported, comprising 15 failed intubations (1:885 general anaesthetics, Table 2), 10 difficult intubations, 10 severe airway problems resulting in desaturation (e.g. severe bronchospasm or laryngeal spasm) and three cases of anaphylaxis.

- Two of the 38 women (one failed intubation and one severe bronchospasm) suffered what were described as hypoxic cardiac arrests. Both made a full recovery after successful intraoperative resuscitation and a period of intensive care.
- Two of the failed intubations occurred following a failed regional technique. In the first of these, a failed spinal was converted to epidural anaesthesia but poor intraoperative anaesthesia resulted in induction of general anaesthesia, failed intubation followed and a laryngeal mask airway was used to continue the general anaesthesia. In the second, an epidural anaesthetic proved insufficient after surgery started and general anaesthesia was induced – failed intubation occurred and a spinal was then used successfully.
- One woman who suffered failed intubation was eventually intubated by a more experienced anaesthetist: following failed intubation for an emergency general anaesthetic for a prolapsed cord ventilation proved impossible; transtracheal jet ventilation was considered but cardiac arrest occurred; cricothyrotomy was about to be performed when more experienced help arrived; the oesophagus was then deliberately intubated and following this tracheal intubation was successful. This patient was resuscitated and, after surgery, she went to ITU for ventilation. In ITU a formal tracheostomy was performed. Despite her cardiac arrest she made a full recovery.

There was one case of awareness with recall in association with general anaesthesia.

DISCUSSION

The three years covered retrospectively in the previous survey (1982, 1987, 1992)³ demonstrated that many UK units were unable to provide data on the anaesthetic techniques used for caesarean section. In 1982 only 36 units could provide complete data, rising to 64 in 1992. The present response from 120 units (approximately 50% of units) is almost double the latter figure. Because of its small response rate, the previous survey³ was open to the criticism 'that the non-responders are at the extremes of the population samples'² and, as was stated during a debate on obstetric anaesthesia, the data may reflect the practice of selected high-performing units while the majority of anaesthetists continue to 'pass gas' for caesarean section. Although it could still be argued that the 50% non-responders in the current survey reflect the extreme and continue to 'pass gas' we do not believe this is so for the following reasons:

1. Calculations based on the number of deliveries per annum and the caesarean section rate^{7,8} indicate that our survey, including as it does 50% of the units in the UK, also includes marginally over 50% of all caesarean sections. This implies that similar sizes and numbers of hospital are included among the responding and non-responding units. The authors are also aware of several units with high regional anaesthesia rates who did not collect data.
2. The regional anaesthesia rate for caesarean section obtained from the official Hospital Episode Statistics for 1994⁸ lies precisely on the line joining the median points for regional anaesthesia use in 1992 and 1997 (Fig. 4), again suggesting a close relationship between the survey data and official national data.

Thus we believe that our data are representative of current practice throughout the UK.

The data suggest that a regional technique is the most common method of anaesthesia for caesarean section and, compared to the previous survey,³ the proportion of regional anaesthetics is increasing (Fig. 4). In 1982 Davis⁹ suggested that 78% of caesarean sections could be performed with epidural anaesthesia and, by the judicious use of spinal anaesthesia, an 84% regional anaesthesia rate was attainable (it should be noted that spinal anaesthesia for caesarean section in the UK was in its infancy at this time). An

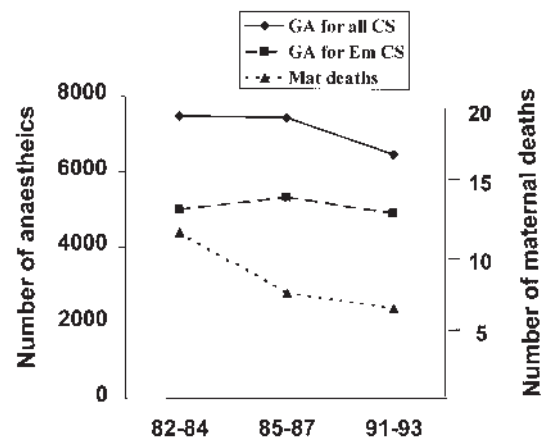


Fig. 6 Changes in the absolute numbers of general anaesthetics given for caesarean section in 31 units in the UK between 1982 and 1992 and the total number of direct anaesthetic maternal deaths in each of the triennial reports.^{3-5,7} GA = general anaesthesia; CS = caesarean section; Em CS = emergency caesarean section; direct anaesthetic deaths related to caesarean section.

accompanying editorial raised the question 'Are his suggestions those of a pie-in-the-sky enthusiast?'¹⁰ and listed the reasons why such a target was scarcely possible. The median rate of 78% in this current survey indicates that many units have more than achieved the regional anaesthesia targets indicated by Davis.⁹ The current maximum regional anaesthesia use for caesarean section was 95% but 10% of the units achieved $\geq 91\%$ regional anaesthesia. These data are remarkably similar to figures reported from Basle (regional anaesthesia rate 89%)¹¹ and the USA,¹² where the median regional anaesthesia rates were 78–88% depending on the hospital group, and in one unit in the USA a regional anaesthesia rate of 96% is reported.¹³ By contrast, these figures are very different from recent publications on anaesthesia for caesarean section in Germany where general anaesthesia rates are in the region of 60–65% for elective caesarean section and 83–98% for nonelective cases.¹⁴⁻¹⁶ It is thought these latter high general anaesthesia rates reflect both the conservative attitude of German anaesthetists and the developmental history of independent departments of anaesthesia in Germany since 1950.¹⁶ Consequently, German women lack information on anaesthesia for caesarean section¹⁶ and are often given no choice (personal communications). This is very different from our own experience and that of others¹³ where women are made very much aware of all the anaesthetic options.

Since 1982 there has been a significant change in obstetric anaesthetic practice and rather than the judicious use of spinal anaesthesia as suggested by Davis,⁹

spinal anaesthesia, in one form or another, now accounts for 56% of all caesarean sections (83% of elective sections, 38% of emergency sections). Despite the widespread literature on CSE anaesthesia the technique accounts for only around 15% of the spinal anaesthetics. The CSE to general anaesthesia conversion rate is about half that of spinal anaesthesia (Table 1). For elective caesarean section the CSE and spinal conversions to general anaesthesia were 0.5% and 1% respectively. Theoretically, the failure rate of CSE is calculated as the product of the individual failure rates for spinal and epidural anaesthesia.¹⁷ The 0.5% conversion rate observed is some 15 times higher than theory predicts¹⁷ and suggests that failure of the spinal and epidural components of a CSE should not be viewed as independent.

In the 1970s, obstetric anaesthesia was the third most common cause of direct maternal death and the vast majority of these deaths followed general anaesthesia for caesarean section, particularly emergency caesarean section. It was a widely held belief that the only way to avoid anaesthetic-related maternal death was to avoid general anaesthesia.¹⁸ Over the past three decades there has been a considerable reduction in anaesthetic-related maternal mortality^{1,4-7,19} and many believe this is linked to the reduction in the use of general anaesthesia.^{1,2} It would appear that such an explanation is too simplistic. Within the hospitals included in the previous survey³ it was demonstrated that, despite a significant reduction in the percentage use of general anaesthesia, because of a greatly increased number of caesarean sections being performed, there was not a significant reduction in the actual number of general anaesthetics used (Fig. 6). Calculations based on official UK data^{8,19} and current and past UK obstetric anaesthesia practices²⁰⁻²² indicate that very similar numbers of general anaesthetics were given for caesarean sections in the 1970-72¹⁹ and 1994-96⁷ triennia (approximately 100 000), yet there were 19 anaesthetic deaths due to general anaesthesia for caesarean section in the 1970-72 triennium¹⁹ and none in the 1994-96 triennium⁷ (the one anaesthetic death in this triennium was related to CSE).⁷ This observation that, in the UK as a whole, there were broadly similar numbers of general anaesthetic caesarean sections in the 1970s and the 1990s negates the often heard statement that the current high regional anaesthesia rate for caesarean section deprives today's anaesthetic trainees of general anaesthesia experience. This may be true for trainees in a few units with very high regional anaesthesia rates but, for the country as a whole, the above data do not support that argument.

While there may be doubt about the contribution made by regional anaesthesia to the reduction in maternal mortality, the data do provide some evidence

on which to assess the potential risks of regional anaesthesia or general anaesthesia in the causation of life-threatening problems. The 38 reported serious complications of general anaesthesia were all potentially life-threatening whereas the 17 reported serious complications associated with regional anaesthesia were readily handled by inducing general anaesthesia (though it is difficult to accept that inducing general anaesthesia for maternal bradycardia is appropriate). Simple mathematics suggest that the risk of a serious life-threatening complication with general anaesthesia occurs in approximately 1:350 general anaesthetics (38 out of 13 275). On the other hand, the 17 serious problems associated with 46 385 regional anaesthetics (1:2728), all of which were managed by the induction of general anaesthesia, would only pose a serious threat to life if the subsequent general anaesthesia caused a problem. Thus a life-threatening problem due to a serious complication of regional anaesthesia would occur in $1:2728 \times 1:350$ cases – i.e. very close to one in a million. This is a very low risk but nevertheless it is a real risk and has been the reason for at least one maternal death.¹ However, this is an underestimate of the true risk associated with regional anaesthesia: as well as the 17 serious complications handled by conversion to general anaesthesia, a considerable number of other patients had regional anaesthesia converted to general anaesthesia. Out of the 48 505 patients where regional anaesthesia was the first choice technique 1410 required conversion to general anaesthesia, i.e. 1:34 of regional anaesthetics (3%) were converted to a general anaesthetic. Thus the overall risk of an attempted regional anaesthetic posing a serious threat to life then becomes $1:34 \times 1:350$ or 1:11 900. This estimate is borne out by data in the current survey: there were four cases reported where the primary regional anaesthesia failed and the subsequent general anaesthesia posed a threat to life (4:48 505; two failed intubations, two suxamethonium anaphylaxis). Thus, the risk of a severe life-threatening complication with regional anaesthesia is closely related to the risk from general anaesthesia multiplied by the regional anaesthesia to general anaesthesia conversion rate. The most significant variable in this equation is the regional anaesthesia to general anaesthesia conversion rate.

Generally, information on complications of anaesthesia, such as general anaesthesia to regional anaesthesia conversion rates, cannot be obtained from official hospital data systems and some systems were even quite inaccurate in classifying elective or non elective procedures. For instance, at the extreme was one hospital with regular elective caesarean section lists: all patients on one of these scheduled lists were classified as elective even if the patient was actually a

dire emergency transferred urgently from labour ward. A true elective case, delayed by such an acute emergency, who then went on to have her surgery outside the normal scheduled elective list was classified as an emergency. It is thus imperative that anaesthetists either collect their own data or have a significant input into those collected by official systems so that the data truly reflect practice.

The reasons underlying high regional anaesthesia to general anaesthesia conversion rates cannot be elucidated from the data but they do not appear to be related to lack of experience with a particular procedure. Excluding units with fewer than 50 of a specific procedure makes no significant difference to the range of conversion rates. Issues that may influence regional anaesthesia conversion rates include the availability of a dedicated obstetric anaesthetist, the experience level of resident obstetric anaesthetists, the ready availability of consultant backup for trainee anaesthetists, the actual practical technique used for epidural, spinal or CSE anaesthesia, the drug regimens used and, most importantly, how the block is assessed. The high intraoperative conversion rates observed must reflect inadequate preoperative assessment of the block. Most anaesthetists continue to use cold or sharp pin prick sensations to test their blocks²³ despite the fact that these sensations bear little relationship to the adequacy of surgical anaesthesia.²⁴⁻²⁶ Although there is no set standard as to what might constitute an acceptable rate of regional anaesthesia to general anaesthesia conversion the knowledge that conversion rates of 1% or less are attainable for all regional anaesthesia techniques should prompt individual units to examine their practice. Herein lies considerable scope for yet further significant reductions in the use of general anaesthesia.

REFERENCES

1. Department of Health, Welsh Office, Scottish Home and Health Department, Department of Health and Social Services, Northern Ireland. Report on Confidential Enquiries into Maternal Deaths in the United Kingdom 1985-1987. London, HMSO 1991.
2. Thomas TA. Views from the Standing Joint Committee of the RCOG and RCA. Royal College of Anaesthetists News Letter 1997; May: 12-13.
3. Brown GW, Russell IF. A survey of anaesthesia for caesarean section. *International Journal of Obstetric Anesthesia* 1995; 4: 214-218.
4. Department of Health Report on Health and Social Subjects 34. Report on Confidential Enquiries into Maternal Deaths in England and Wales 1982-84. London, HMSO 1989.
5. Department of Health, Welsh Office, Scottish Office Home and Health Department, Department of Health and Social Security, Northern Ireland Report on Confidential Enquiries into Maternal Deaths in the United Kingdom 1988-1990. London, HMSO 1994.
6. Department of Health, Welsh Office, Scottish Office Home and Health Department, Department of Health and Social Services, Northern Ireland Report on Confidential Enquiries into Maternal Deaths in the United Kingdom 1991-1993. London, HMSO 1996.
7. Department of Health, Welsh Office, Scottish Office Department of Health, Department of Health and Social Services, Northern Ireland. Why Mothers Die. Report on Confidential Enquiries into Maternal Deaths in the United Kingdom 1994-1996. London, The Stationery Office 1998.
8. NHS Maternity Statistics, England: 1989-90 to 1994-95. Department of Health Statistical Bulletin 1997/28.
9. Davis AG. Anaesthesia for caesarean section. The potential for regional block. *Anaesthesia* 1982; 37: 748-753.
10. Corral I. Editorial. *Anaesthesia* 1982; 37: 711-713.
11. Schneider MC. General anaesthesia for caesarean section revisited. In: Bogod DG, ed. *Obstetric Anaesthesia*, Baillieres Clinical Anaesthesiology 1995; 9: 649-673.
12. Hawkins JL, Gibbs CP, Orleans M, Martin-Salvaj G, Beaty B. Obstetric anaesthesia work force survey, 1981 versus 1992. *Anesthesiology* 1997; 87: 135-143.
13. Tsen LC, Pitner R, Camann WR. General anaesthesia for cesarean section at a tertiary care hospital 1990-1995: indications and implications. *International Journal of Obstetric Anesthesia* 1998; 7: 147-152.
14. Josten KU, Wolf H, Schäfer RD. Obstetric Anesthesia 1988-1996 in Northrhine/Germany: Results of the perinatal survey at the Chambers of Physicians. *Anesthesiology* 1998; 89: 1288-1290.
15. Meuser T, Eichler F, Grond S, Winkler B, Lehmann KA. Anästhesieverfahren zur sectio caesarea in Deutschland. *Anaesthesist* 1998; 47: 557-564.
16. Stamer UM, Messerschmidt A, Wulf H. Anaesthesia for caesarean section - a German survey. *Acta Anaesthesiol Scand* 1998; 42: 678-684.
17. Lyons G. Controversies in obstetric anaesthesia: Epidural is an outmoded form of regional anaesthesia for elective caesarean section. *International Journal of Obstetric Anesthesia* 1995; 4: 34-36.
18. Moir DD. *Obstetric Anaesthesia and Analgesia*, Baillière Tindall, London, 1980; pp 145.
19. Department of Health and Social Security Report on Health and Social Subjects 11. Report on Confidential Enquiries into Maternal Deaths in England and Wales 1970-1972. London, HMSO 1975.
20. Tunstall ME. Complications of epidural analgesia. In: Doughty A. (Ed) *Proceedings of the Symposium on Epidural Analgesia*. London: HK Lewis, 1972; pp 64-66.
21. Moir DD. Indications for epidural analgesia with special reference to the management of pre-eclampsia. In: Doughty A. (Ed) *Proceedings of the Symposium on Epidural Analgesia*. London: HK Lewis, 1972; pp 73-79.
22. Crawford JS. Observations on one thousand epidural blocks given in labour. In: Doughty A. (Ed) *Proceedings of the symposium on Epidural Analgesia*. London: HK Lewis, 1972 pp 83-88.
23. Bourne TM, deMelo AE, Bastianpillai BA, May AE. A survey of how British obstetric anaesthetists test regional anaesthesia before Caesarean section. *Anaesthesia* 1997; 52: 901-903.
24. Russell IF. Levels of anaesthesia and intraoperative pain at caesarean section under regional block. *International Journal of Obstetric Anesthesia* 1995; 4: 71-77.
25. Rocco AG, Raymond SA, Murray E, Dhingra U, Freiburger D. Differential blockade of touch, cold, and pinprick during spinal anaesthesia. *Anesth Analg* 1985; 64: 917-923.
26. Liu S, Kopacz DJ, Carpenter RL. Quantitative assessment of differential sensory nerve block after lidocaine spinal anaesthesia. *Anesthesiology* 1995; 82: 60-63.

A section of the form for data collection for a UK
National Hospital Survey of Anaesthesia for Caesarean Section
1997

[illegible]